

The Chinese University of Hong Kong, Shenzhen

Course Outline

1. Course identity

A. Course as listed at CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	MFE5160
Course title (English)	Quantitative Risk Management
Course title (Chinese)	定量风险管理
Units	3
Description (English)	This course provides students with an in-depth understanding of financial risks by building mathematical models in order to understand the nature of these risks and to manage them using relevant products and instruments. Topics include market risk such as interest rate risk, credit risk, foreign exchange risk at trade level and portfolio level. The course also covers Value at Risk, liquidity risk and an introduction to bank regulations.
Description (Chinese)	本课程为学生对金融风险提供一个深入了解,通过建立数学模型,以明了这些风险的性质,去量度及使用相关产品和工具来管理它们。教学主题包括在交易层面和投资组合层面的市场风险,如利率风险,信用风险,外汇风险。本课程还包括风险价值,压力测试,流量风险,操作风险,并介绍现今市场监控银行的条例。

B. Corresponding course at CUHK

Please give details of the closest corresponding course at CUHK (as approved by CUHK Senate and listed in course list). If the course in Shenzhen maps to more than one course at CUHK, please make multiple copies of the block below.

Course code	
Course title (English)	
Course title (Chinese)	
Units	
Description (English)	
Description (Chinese)	

2. Course requisites

Please state prerequisites and co-requisites, in terms of courses at CUHK-Shenzhen or any other requirements.

A. Prerequisites

The pre-requisite course is Financial Derivatives (MFE5130) or some courses of equivalent level. Students need to have some knowledge of derivatives products. To help understand the materials, students should have a general understanding of the global financial markets.

B. Co-requisites

No co-requisite.

C. Exclusion

D. Others

3. Learning outcomes

Upon completing the course, the student will be able to:

- understand the objectives of risk management and the trade-offs between risk and return
- apply the principles and tools for measuring and managing risk in order to develop hedging strategies
- assess the limitations of standard methodology in risk management
- follow the market development of risk management, current regulations and industry standard

4. Course syllabus

Please highlight fundamental concepts involved in each topic to help students better understand what is covered in the course.

- Introduction to risk management
- Risk vs return
- Mean, variance, skew and kurtosis
- Forwards, futures, swaps, options and other derivatives tools
- Value at Risk (VaR), expected shortfall, extreme value theory
- Interest rate risk, yield curve construction
- Foreign exchange risk, covered interest arbitrage

- Credit risk, default probability, Merton model, Credit/Debit Value Adjustment (CVA/DVA)
- Liquidity risk, liquidity adjusted VaR, optimal execution
- Operational risk, basic indicator, standardized & Advanced Measurement Approach (AMA)
- Greeks: delta, gamma, vega, theta, rho and cross greeks
- Stress testing, scenario analysis, stressed VaR
- Monte Carlo simulation
- Basel I, II, III
- Dodd-Frank Act
- Volcker rule
- Sarbanes-Oxley Act

5. Assessment scheme

Component/ method	% weight
Assignments	20
Project	35
Final Exam	45

Remarks:

6. Grade descriptor

The grade descriptor should align with specific course learning outcomes, not a generic description applicable to all courses.

Type:**General**

Grade	Description
A	- Demonstrate exceptional understanding of the course materials in assignments, project and final.
A-	- Demonstrate solid understanding of the course materials in assignments, project and final.
B	- Demonstrate substantial understanding of the course materials in assignments, project and final.
C	- Demonstrate fair understanding of the course materials in assignments, project and final.
D	- Demonstrate limited understanding of the course materials in assignments, project and final.
F	- Demonstrate unsatisfactory understanding of the course materials in assignments, project and final.

7. Feedback for evaluation

Please specify forms of evaluation to generate feedback on classes from students.

Formal Course and Teaching Evaluation

Informal feedback to instructor and/or teaching assistant

Departmental retreat and programme review

8. Readings

Required:

1. Risk Management and Financial Institutions, 6th Edition, John Hull, 6, United States

Others:

9. Course components

Activity	Hours/week
Lecture	3

10. Indicative teaching plan

Week	Content/ topic/ activity
1	• Introduction to Risk Management • Securitization and the Global Financial Crisis • Volatility
2	• Correlation and Copulas • Valuation and Scenario Analysis
3	Value at Risk & Expected Shortfall
4	Historical Simulation and Extreme Value Theory
5	Model-Building Approach
6	Interest Rate Risk
7	Derivative Risk
8	• Scenario Analysis and Stress Testing • Estimating Default Probabilities

9	xVAs
10	Credit Value at Risk
11	• Operational Risk • Liquidity Risk
12	Basel I, II, and Solvency II
13	Basel II.5, Basel III and Other Post-Crisis Changes
14	Final Examination

11. Implementation plan

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections. Example: 150 students for lecture (x 2); 30 students for tutorials (x 10).

12. Academic honesty and plagiarism

A course outline may include subject-specific requirements on plagiarism, in addition to the University established policies.

Students must complete all coursework and assessments independently, unless AI tool use is explicitly permitted. Unauthorized or improper use of AI may constitute academic dishonesty and harm learning. Misuse includes submitting AI-generated work as one's own, using AI to cheat, or employing AI unethically. Always obtain permission before using AI in academic work, and be sure to acknowledge any AI use in submissions.

13. Approval

Has the course title been included in the programme submission approved by CUHK Senate? Are there any differences? Have the details (as in this document) been approved at School or other level in CUHK- Shenzhen?

Submit application

14. Any other information

Honesty in Academic Work

The Chinese University of Hong Kong, Shenzhen places very high importance on honesty in academic work submitted by students, and adopts a policy of zero tolerance on academic dishonesty. Any related offence will lead to disciplinary action including termination of studies at the University. (see <https://registry.cuhk.edu.cn/page/30>).

15. Version date

Version number	1
As of (date)	2025-10-31